

QQ Super Compression

User Manual / 1.2.0

Control dynamics. Keep the source's attack.

For vocals, instruments, buses and mastering. There are no conventional Attack / Release controls: shape dynamics with Lookahead, Ratio, Threshold and Mix.

Single can control dynamics downward or boost within a selected range. Dual provides separate upward and downward controls in the same interface.



Start with what you hear

Decide how steady the dynamics should be and how much articulation to retain. The Display helps explain changes; it does not replace listening.

Quick start and navigation

A first session in six steps.

- 1 Insert on the track to process**
 Start with SINGLE, ST and SC: INT so the plugin follows the track itself.
- 2 Choose Lookahead**
 26 ms is a practical starting point, not a rule for every source.
- 3 Start at 1:1**
 All Ratios default to 1:1, with no dynamic change. To try downward processing, raise Ratio and leave Range OFF. For upward and Dual processing, see pages 7-9.
- 4 Use Mix to set the processing amount**
 If processing is too strong, lower Mix to blend back the original dynamics. Both boost and cut displays follow Mix.
- 5 Compensate level and compare**
 Play a representative passage, use Match, then trim Makeup / Output Gain to avoid loudness bias.
- 6 Listen in the full mix**
 Use Bypass or A/B to judge articulation, consistency and groove - not only the GR number.

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Know the interface

The same audio controls remain in place across all themes.



Light / ST. The values illustrate the interface; they are not a recommended preset.

Area	Use it to
Display	Compare original dynamics, boost / cut including Mix, and final output.
Two boundary faders	Threshold / Range in Single; UP / DOWN thresholds in Dual.
Input / Output / Gain +/-	Check input, output and effective dynamic gain.
Bottom knobs and SINGLE / DUAL	Adjust gain, Ratio and Mix; switch Single / Dual beside Ratio.
Mode / LINK / Monitor	Choose channel processing, link controls or audition domains.
Top right	Open sidechain, change theme, use A/B and Bypass.

The interface uses a 3:2 landscape aspect ratio. Drag the bottom-right corner to resize it. The plugin remembers the last size when reopened.

Themes and everyday controls

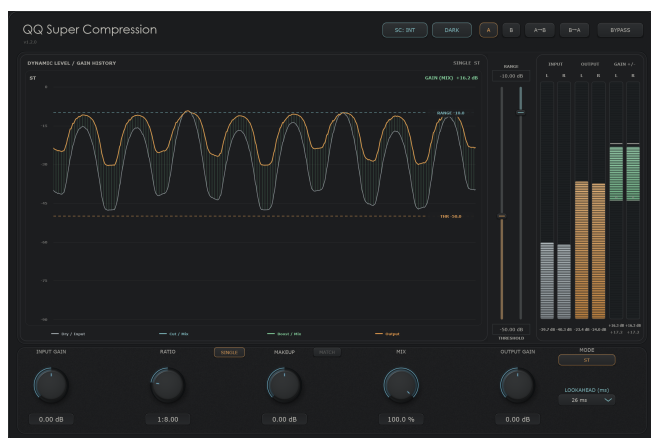
Click the top-right button labelled LIGHT, DARK or CLASSIC to cycle themes. The last selection is restored when the plug-in opens again.



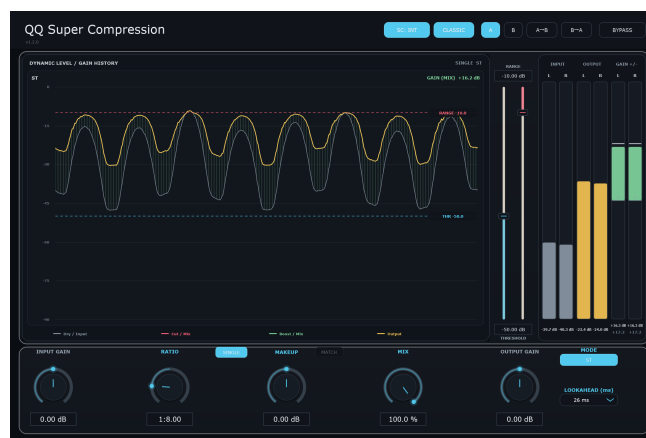
The theme button is between SC and A/B.

Theme	Appearance
Light	Light interface.
Dark	Subdued dark interface.
Classic	Original classic dark interface.

All themes share the same layout, parameters and sound. Choose the appearance that suits your working environment.



Dark



Classic

Gesture	Result
Drag / Shift-drag	Normal / fine adjustment.
Double-click a value	Type an exact value.
Alt + left click	Restore the default. When linked, related controls follow within their allowed ranges.
Ctrl/Cmd + Z	Undo; add Shift to redo.

Ratio, Input and Mix

Three useful controls, not interchangeable settings.



Ratio

Single centers on 1:1: above it, processing is downward; below it, processing is upward. Within the active interval, larger downward values generally mean more cut, and smaller upward values mean more boost. Dual provides separate UP RATIO and DOWN RATIO controls; see page 8.

Ratio here is not the fixed dB slope above threshold used by a conventional compressor. Do not assume the same values will give the same result. All Ratios default to 1:1.

Input Gain

Input Gain changes the level entering the processing path, including the Dry path. With INT it also changes the detector level relative to Threshold. With EXT, the external send and Key Gain set the detector level; Input Gain is not a substitute.

Mix

Mix blends Dry with processed Wet. It is also a useful processing amount control: first make the boost or cut clear, then lower Mix to blend back the original dynamics you want to retain.

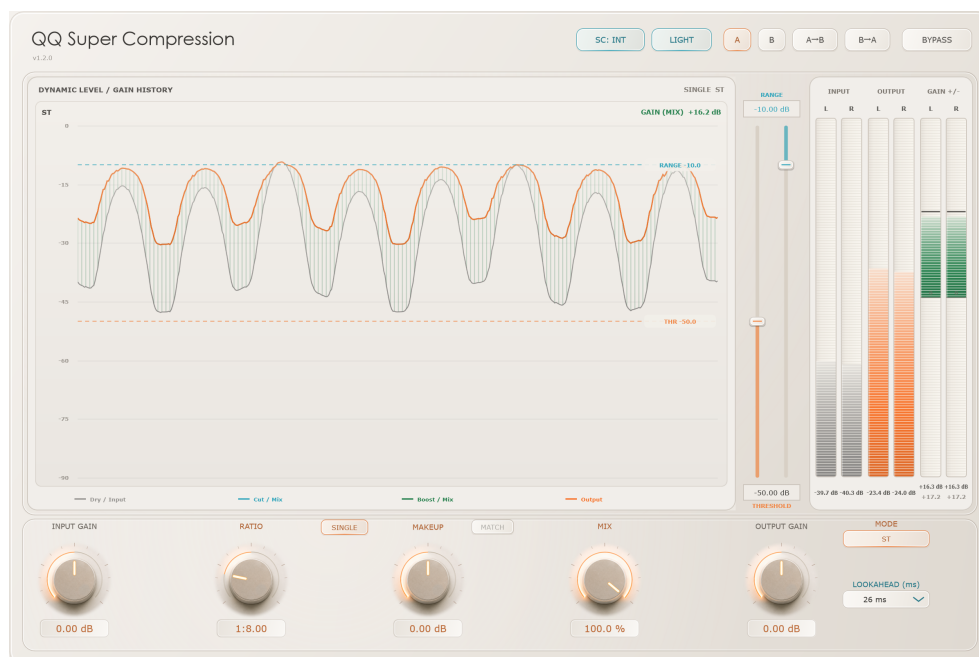
Dynamic gain includes Mix

At Mix = 0%, effective dynamic gain is 0 dB; at 100%, the full processing amount is retained. Intermediate values are not simply the gain in dB multiplied by Mix. Makeup and Output Gain are excluded from this dynamic gain.

A practical order: find the compression character with Ratio, set its final strength with Mix, adjust Input Gain when you need a different input level, then compare loudness again.

Single: Threshold and Range

Choose the active levels, then set direction and strength with Ratio.



SINGLE: the lower Threshold and upper Range define the processing interval. The faders align with the dashed lines in the display.

Detector level	Dynamic processing
At or below Threshold	No dynamic boost or attenuation.
Above Threshold and below Range	Upward or downward processing, according to Ratio.
At or above a finite Range	Returns fully to 0 dB dynamic gain.

Range OFF: no upper cutoff

At its top OFF position, Range imposes no upper limit and retains the original downward behavior. Pull it down to enable a finite cutoff. A finite 0 dB setting is different from OFF. Threshold OFF means there is no lower gate.

When the faders meet

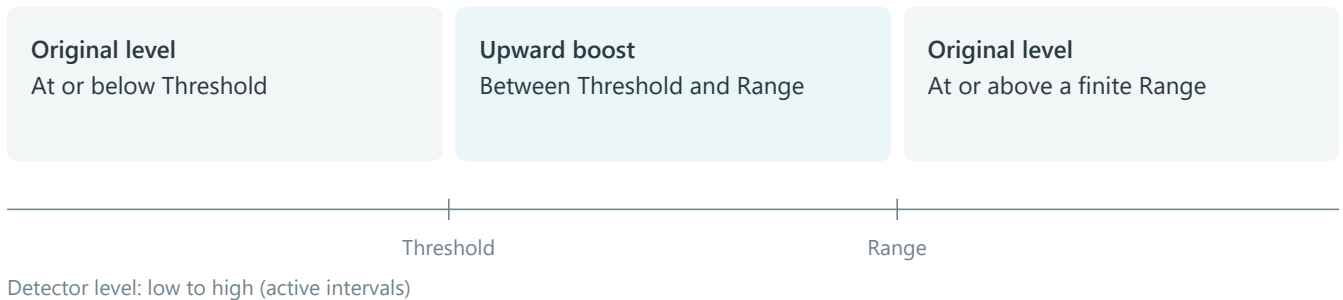
Range stays at or above Threshold. When one fader reaches the other, it moves the other along; coincident boundaries disable dynamic processing. Drag back to separate them.

Here, 0 dB dynamic gain means no dynamic boost or cut. Input, Makeup, Mix and Output Gain still follow their settings. In EXT, these boundaries apply to the external detector signal.

Upward: boost above the gate

Bring selected detail forward using Super Compression's dynamic processing approach.

In SINGLE, move Ratio below 1:1, for example to 1:2, 1:4 or 1:8, to enter upward processing. The displayed 1:8 means 1/8. Smaller values generally give eligible material more boost.



The gate enables boost; it does not mute audio

Threshold acts as a gate that allows boost above it. Below the gate, the original level is retained: quieter sounds are neither muted nor specifically boosted.

1 Select the detail to bring forward

Select SINGLE. Set a lower Threshold and upper Range, for example -50 dB / -10 dB. Listen for the detail you want between them. These values are examples only.

2 Then adjust Ratio and Mix

Move gradually from 1:1 toward 1:2 and 1:4. If the boost is too strong, reduce its strength or lower Mix. Do not substitute Makeup for dynamic boost.

3 Listen to quiet and loud passages

Quieter parts within the active interval generally receive more boost. Boost approaches 0 dB near the upper boundary. Range OFF removes the finite upper cutoff.

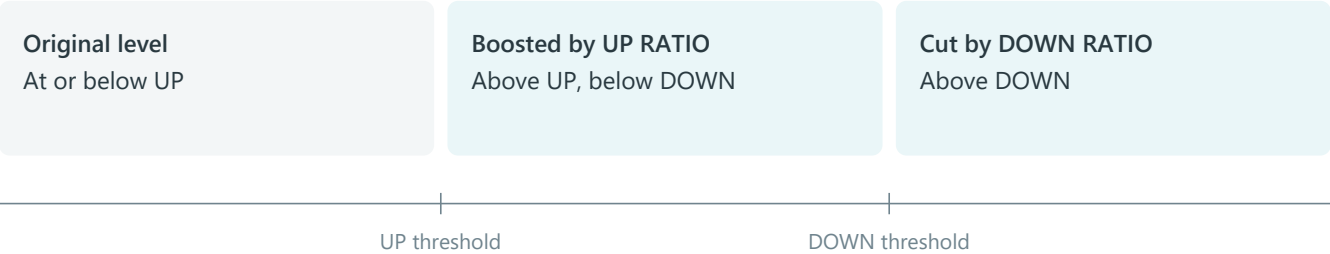
To leave very quiet breaths, noise or reverb at their original level, place the lower gate above them. Check that the desired material still enters the boost interval. Selection follows detector level, not sound type.

Dual: lift quieter parts, control louder ones

Click the small button beside Ratio to change SINGLE to DUAL.



DUAL: UP RATIO and DOWN RATIO control the two directions separately. UP / DOWN thresholds replace Threshold / Range.



At DOWN, dynamic gain returns to 0 dB. Above it, upward processing stops and only Down Ratio applies. The two gains never stack at the same detector level. Dual has no Range control.

Starting a new instance	How to read it
UP = -inf; DOWN = 0 dB	The lower UP gate is fully open; upward processing is not disabled.
Up Ratio = 1:1; Down Ratio = 1:1	Both directions are neutral. Adjust a Ratio to hear processing.

Threshold order and collision
DOWN must stay at or above UP. When one fader reaches the other, it pushes the other along. Dynamic processing stops when the thresholds coincide.

Ratio LINK: opposite proportional changes

The small LINK above and between the two Ratios appears only in DUAL.



Up and Down retain their existing relative relationship. Enabling LINK does not make values jump.

Ratio LINK defaults to on at first use, then remembers your last choice. When reopening a saved project, the LINK state saved in that project takes priority.

Initial Down / Up	Adjust Down	Linked Up
1:1 / 1:1	1:1 → 2:1	1:2
4:1 / 1:2	4:1 → 8:1	1:4
8:1 / 1:4	8:1 → 4:1	1:2

Preserve the relationship, without re-pairing

For example, doubling Down halves the Up value; adjusting Up uses the same inverse relationship. Starting at 4:1 and 1:2 will not force them into a reciprocal pair. Dragging, fine adjustment and direct entry all follow this rule.

At a limit, both stop together

Up ranges from 1:32 to 1:1; Down from 1:1 to 32:1. If either would exceed its range, both stop at the shared limit. Reverse the adjustment to continue. Toggling LINK alone does not change the Ratios.

To hear upward processing alone

Turn off the small Ratio LINK, leave Down Ratio at 1:1, then adjust Up Ratio alone. With LINK on, changing Up also changes Down, so you hear both settings change together.

Also available in LR / MS

The small LINK scales down with compact layouts and stays between Up / Down Ratio. It links those Ratios within each channel. Channel LINK beside Mode links L/R or M/S; see page 15.

Match loudness before comparing

Control	Use
Makeup	Compensates Wet level before Mix. At lower Mix it is not a global output trim.
Output Gain	Changes the final output level without changing the dynamic gain display.
Match	Makes a one-time loudness adjustment to Makeup from played material; it is not continuous auto gain.

- 1 Play representative material**
 Wait until Match is available, then click it. Very short or silent passages may not provide a usable reading.
- 2 Check final output loudness**
 Match compares Dry with Wet before makeup; the final mixed output may still need a small Makeup or Output Gain correction.
- 3 Compare sound, not level**
 Toggle Bypass and listen to articulation, tails and consistency across the passage.

Use A / B for two alternatives

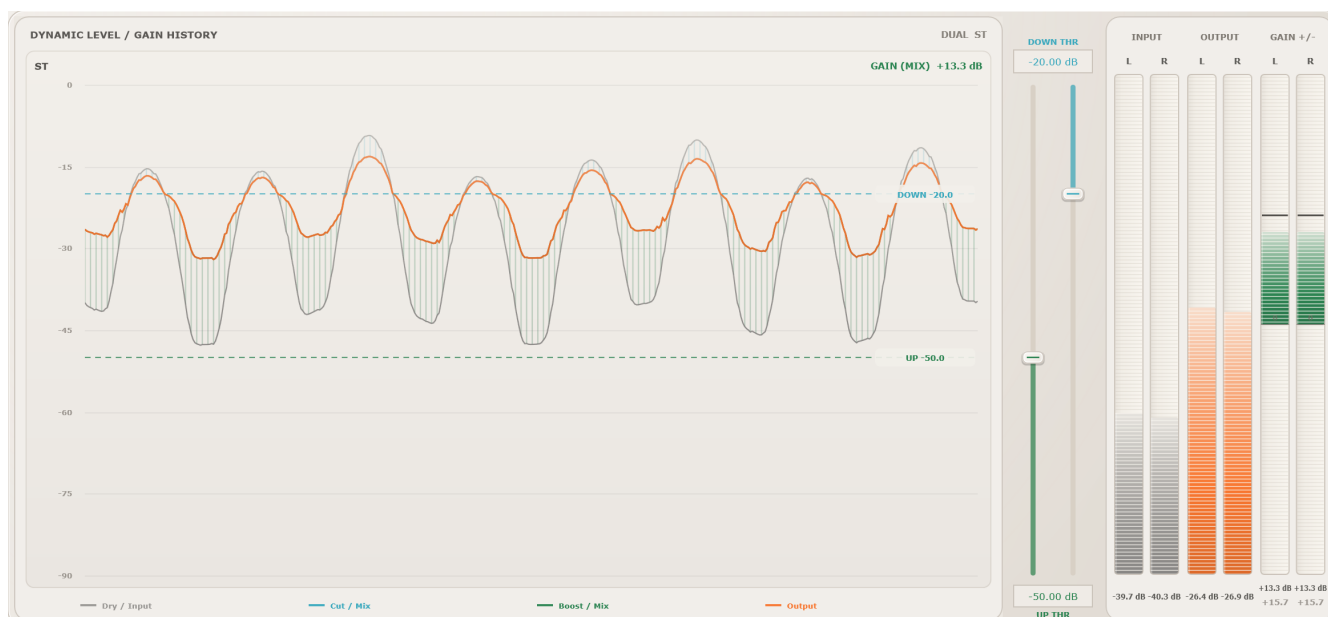
Set up A, copy with A→B, then switch to B and try another approach. Click A or B to compare. B→A copies in the other direction; copying replaces the destination's sound settings.

A/B includes sidechain Source, Key Gain and HPF. Theme is not a sound setting; Bypass, LINK and SC LISTEN are not recalled by the A/B sound snapshot.

0% Mix is not full Bypass

At 0% Mix, Input Gain and Output Gain still apply. Bypass compares the original signal without those gain and compression stages.

Read the Display and meters



In the same history view, green shows boost and Cut / Mix shows attenuation. Orange is the final output.

What you see	Meaning
Dry / Input	Original dynamics before Input Gain; the grey line does not move as a whole when Input changes.
Cut / Mix	Dynamic attenuation including Mix. Its distance from Dry indicates the effective cut.
Boost / Mix	Dynamic boost including Mix, shown in green. Its distance from Dry indicates the amount of boost.
Output	Final output including Makeup, Mix and Output Gain.
External key	When EXT is selected and the external input is available, a lighter trace shows the sidechain reference to help check triggering.
Gain +/-	Zero is at the center: boost above, cut below. Positive values indicate boost and negative values indicate attenuation. Hold retains recent prominent changes.

Dynamic gain excludes Makeup / Output Gain and is 0 dB at Mix = 0%. All three themes show boost and cut; match the colors to the named legend below the graph.

History responds to current settings

Changing Ratio, thresholds, Range or Mix updates the visible history. EXT Key Gain updates immediately; HPF refreshes after you release it and may show HPF UPDATING. The graph is a reference under the current settings, not audio replay.

Lookahead and the 0 ms option

Lookahead is an advance observation window, not Attack. Longer windows generally suit clean, stable dynamic control.

A longer window also increases plugin latency, which the host will usually compensate.



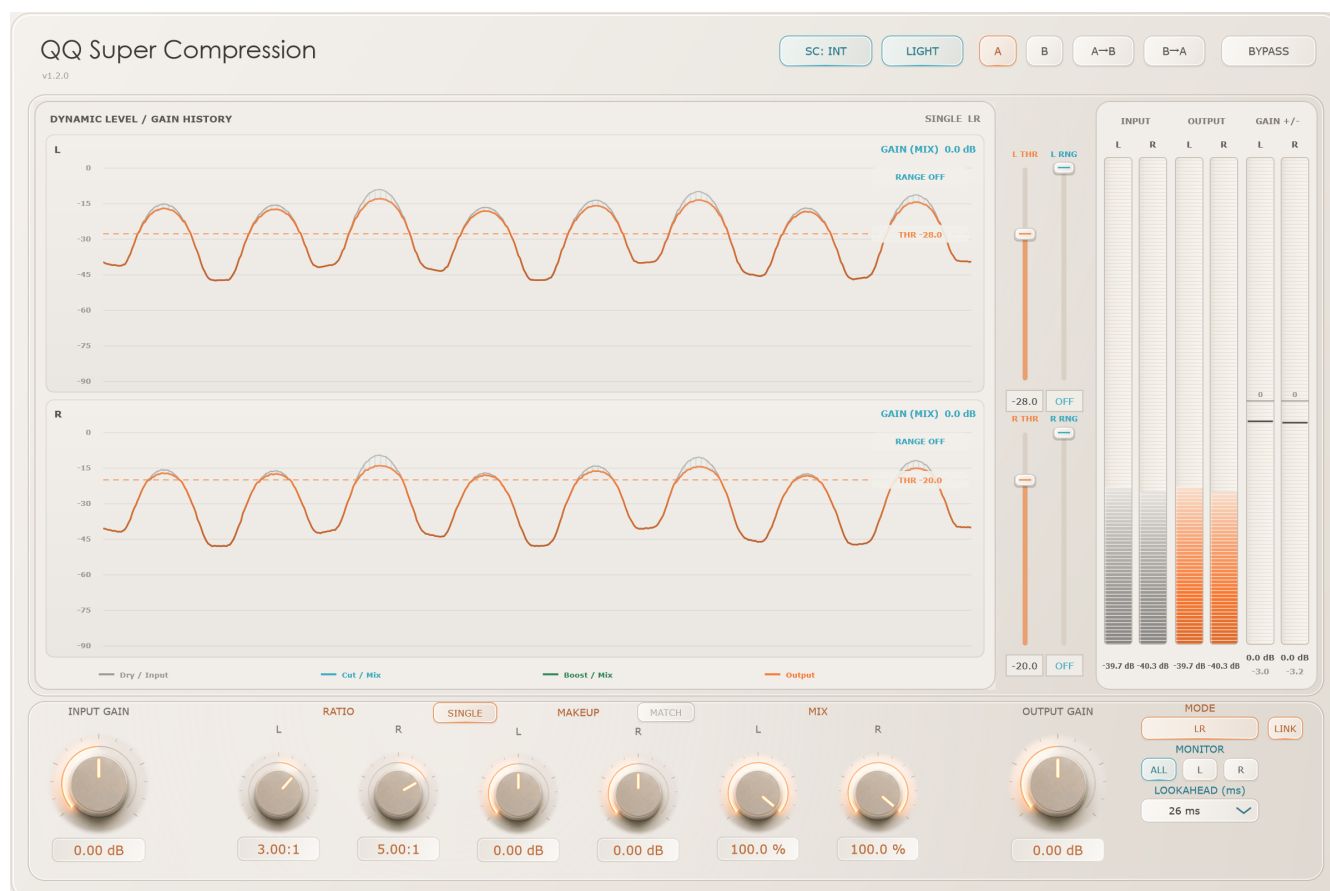
At 0 ms, Oversampling appears below Lookahead.

Choice	Use and trade-off
10 ms	Shorter window and lower latency.
26 / 40 ms	Useful starting points; choose by low-frequency stability and the overall sound.
80 / 100 ms	Longer windows; listen for greater stability and consider the extra latency.
0 ms	More pronounced nonlinear colour, not the most transparent mode.

At 0 ms choose 1x / 8x / 16x Oversampling. Start with 8x; higher rates cost processing and latency. The control is hidden at 10 ms and above. 0 ms with 8x / 16x still adds latency.

ST and LR: together or independently

Click Mode at the lower right to cycle ST → MS → LR. Start with ST; choose LR when the two sides genuinely need different treatment.



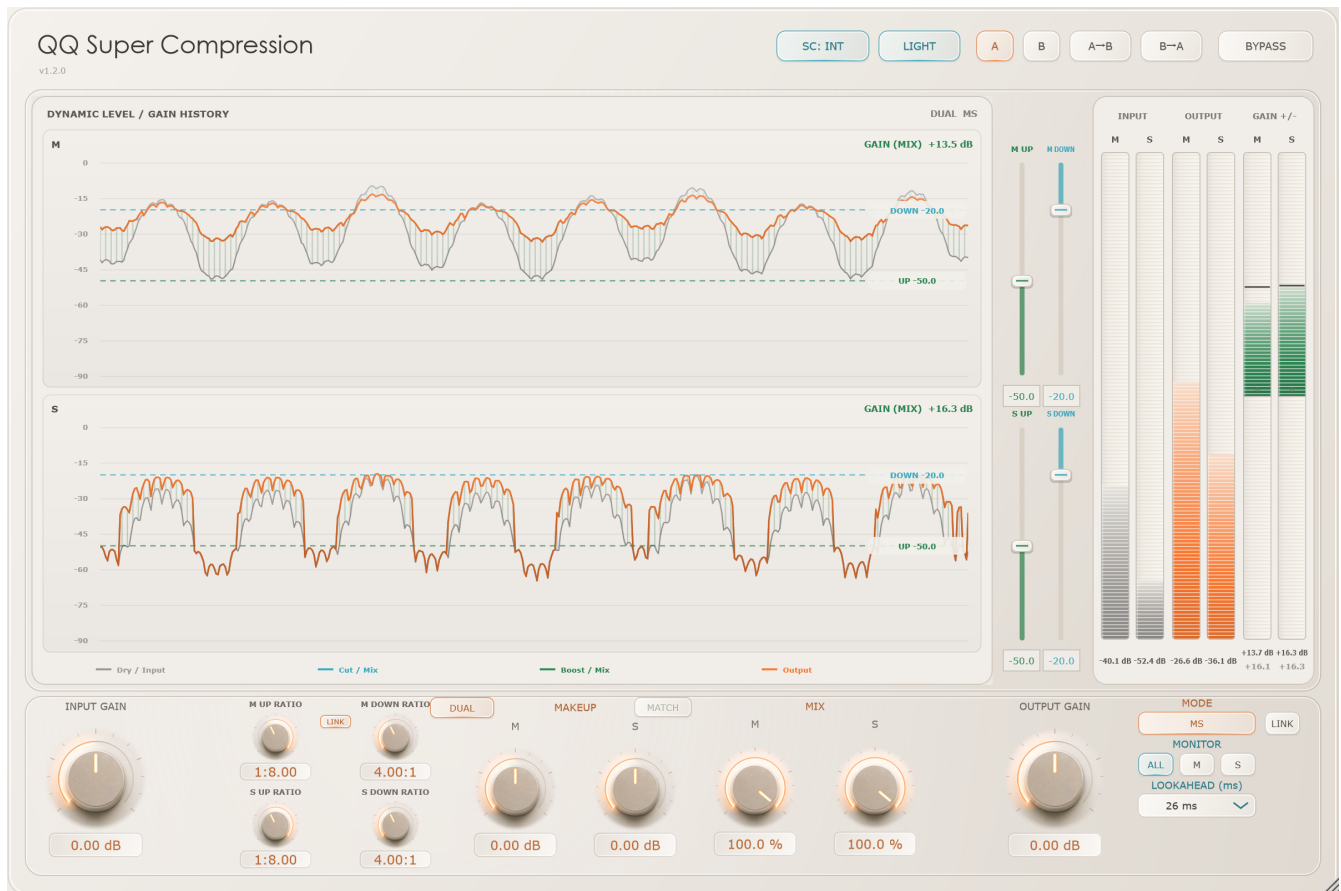
LR: L above, R below, with independent controls.

Mode	How to use it
ST	Uses common compression gain for stereo; a useful starting point for tracks and buses.
LR	Left and right have independent Ratio, boundary, Makeup and Mix controls. In DUAL, each side has its own UP / DOWN. Check for image shifts when processing asymmetric material.

MONITOR

Choose L or R to audition that side centred. Return to ALL to check normal stereo, and do not leave one-sided audition active for the final output.

MS: control centre and sides



MS / DUAL: Mid is above Side. Each channel has its own UP / DOWN Ratios and thresholds.

Mid mainly represents content common to both sides; Side represents their differences. They are not a precise separation of a centre track from every other track.

1 Listen to M and S first

Use M / S under MONITOR, then return to ALL.

2 Adjust each for a reason

For steadier centre dynamics, start with M Ratio / Mix. Treat S cautiously when preserving ambience and width.

Recheck the full mix

Compression or level changes in M/S can alter the sense of space. In ALL, check centre, width and mono compatibility, not only soloed domains.

Channel LINK: adjust together, keep differences

In LR or MS, LINK beside Mode links matching parameters across channels. Set the two sides separately, then enable it to adjust them together without forcing identical values.



SINGLE example: channel LINK beside Mode is on, while the left and right Ratios remain at 3:1 and 5:1.

Parameter	Before	After an equal change
Ratio	3:1 / 5:1	4:1 / 6:1
Threshold	-20 / -10 dB	-18 / -8 dB
Makeup	-3 / +1 dB	-2 / +2 dB
Mix	100% / 70%	90% / 60%

1 Set the relationship with LINK off

Decide which domain needs more compression or a different blend.

2 Enable LINK for joint adjustments

Move either parameter in a pair; the other follows by the same change. Fine adjustment and direct entry work too.

Both stop at a limit

If either side would exceed its range, both stop where the difference can be retained. One Threshold at OFF cannot form a finite dB offset; enable both thresholds first if you want that relationship.

The two LINK controls in DUAL

The small LINK between the Ratios links Up / Down within each channel. LINK beside Mode links matching L/R or M/S parameters. With both enabled, channels move together while each channel retains its existing Up / Down relationship.

Sidechain: start with the panel

Click SC: INT / SC: EXT at the top right. Its label shows the current source.



Control	How to use it
INT / EXT	INT follows the current track itself. EXT uses a signal sent from another track to control boost or attenuation on the current track.
KEY GAIN	Available only with EXT. Adjusts the external detector level and trigger strength, not the main track level.
HPF	Reduces low frequencies in the INT or EXT detector key. OFF is full-band; active range is 20-500 Hz.
KEY LEVEL	Check that the detector receives a signal. With EXT showing N/A, first check that the host sidechain bus is enabled.
SC LISTEN	Temporarily audition the detector key to check routing and HPF. Switch it off before judging normal output.

Two common tasks

Low notes pull the whole part down: try INT + HPF, page 17.

Want bass to duck when the kick plays? Use EXT, page 18.

Internal sidechain: reduce bass dominance

Use this when kick or bass repeatedly pulls down the rest of a drum bus, mix bus or master. The goal is less low-frequency triggering, not less bass in the audible signal.

- 1 Start with INT and hear the problem**
Loop a passage with bass and other content together. Leave HPF OFF to establish a comparison.
- 2 Open the sidechain panel**
Click SC: INT and keep SOURCE on INT. A disabled Key Gain is normal here.
- 3 Raise HPF gradually**
Start at a low cutoff and listen for less whole-mix ducking on bass events. Make small changes.
- 4 Audition the key if needed**
Use SC LISTEN to hear which bass content HPF removes, then turn it off and return to the full sound.
- 5 Inspect history after release**
HPF history refreshes after you release the knob. If HPF UPDATING appears, wait for it to clear before comparing.
- 6 Rebalance the compression amount**
Filtering may reduce GR or change its shape. Revisit Ratio, Threshold or Mix as needed, then compare at similar loudness.

It is normal not to hear bass being cut

HPF filters only the key used to decide compression, not normal output. It is not equivalent to Key Gain: Key Gain scales the detector level; HPF changes frequency weighting. Too high a cutoff can make the compressor ignore bass that still needs control.

External sidechain: kick-controlled bass

Insert the plug-in on the bass - the track to turn down. Send the kick to its sidechain as the trigger. Do not reverse the tracks or confuse a normal audio send with a sidechain send.



- 1 Enable the host sidechain input**
 Enable sidechain input for this plug-in instance on the bass. Add and enable a kick send to that sidechain. Names and locations vary by host.
- 2 Select EXT in the plug-in**
 Open SC, choose EXT and play the kick. Check KEY LEVEL first; if it has no reading, fix routing before increasing Ratio.
- 3 Confirm the actual trigger**
 Briefly enable SC LISTEN: you should hear the kick, not the bass. Turn audition off afterward.
- 4 Set trigger level and reduction**
 For downward ducking, start with SINGLE, Ratio above 1:1 and Range OFF. Set detector strength with Key Gain or the send level, and set the lower boundary with Threshold. Leave HPF OFF initially.
- 5 Use Mix to decide how much room to make**
 Find an audible duck, then reduce Mix to retain the desired bass sustain. Listen to both tracks together, not only solo bass.
- 6 Decide whether HPF helps**
 Raising HPF reduces low-frequency kick triggering and may weaken the duck you want. Use it when you want more emphasis on the attack or less unwanted low-bass triggering.

There are no conventional Attack / Release controls. Ducking shape depends on the trigger, Lookahead and compression settings; it is not guaranteed to be smoother than every conventional sidechain.

Apply it in a real mix

Vocals, guitar, piano: consistency with articulation

Try SINGLE / ST / INT and 26 ms. Raise Ratio above 1:1, then use Mix to keep consonants, plucks or hammer attacks natural. To process only louder parts, enable Threshold and leave Range OFF. Use Match and compare in the mix.

Tails and quiet phrases: bring detail forward

Try Single upward processing to lift quieter words, pluck tails or reverb. Use Threshold to exclude unwanted very quiet sounds and Range to set the upper limit. To also control louder parts, switch to Dual and adjust Up and Down separately. Turn Ratio LINK off when first auditioning each direction.

Mix bus / mastering: a more cohesive balance

When you want a steadier, more cohesive mix without deliberately reshaping drum attacks through conventional Attack / Release, try it as an alternative to G Bus-style compression. Start gently, compare balance across sections and lower Mix when needed.

If every kick pulls the whole mix down, try INT + HPF. Do not raise the cutoff merely to make the GR number smaller. At similar loudness, check drum attack, low end, space and the song's larger dynamic movement.

Vocal-keyed accompaniment: make space for words

Insert the plugin on the backing bus, send the vocal to its external sidechain and choose EXT. Confirm the source with SC LISTEN. Select SINGLE, Ratio above 1:1 and Range OFF, then adjust Key Gain, Threshold and Mix so the backing steps back while the vocal plays.

If plosives trigger too much ducking, try raising sidechain HPF. If breaths or noise also make the accompaniment dip, check the sent signal and Threshold; HPF alone will not solve every unwanted trigger.

For every comparison

Turn SC LISTEN off, return MONITOR to ALL, and compare at similar loudness. Use an appropriate limiter when you need a strict final peak ceiling; this plug-in is not a brick-wall limiter.

Quick reference: dynamics and range

Parameter	Range / choices	Notes
SINGLE / DUAL	Single / Dual	Small button beside Ratio; each mode retains its own parameters.
Single Ratio	1:32 to 32:1	Default 1:1; below 1 is upward, above 1 is downward.
Up Ratio	1:32 to 1:1	Default 1:1; upward boost only.
Down Ratio	1:1 to 32:1	Default 1:1; downward compression only.
Threshold	OFF / -119.99 to 0 dB	Single lower boundary; default OFF.
Range	OFF / -inf to 0 dB	Single upper boundary; default OFF. Finite 0 dB is different from OFF.
UP THR	-inf to 0 dB	Dual lower boundary; default -inf.
DOWN THR	-inf to 0 dB	Dual handoff threshold; default 0 dB.
Ratio LINK	OFF / ON	Initially ON; remembers your choice. Saved project state takes priority.
Mode	ST / MS / LR	Channel processing mode, selected independently of SINGLE / DUAL.
Channel LINK	OFF / ON	Beside Mode; links matching LR / MS parameters while preserving their relative difference.

Boundary order and defaults

In Single, Range stays at or above Threshold; in Dual, DOWN stays at or above UP. Coincident boundaries disable processing. All three Ratio types default to 1:1 on every channel. Existing projects retain their saved values.

1:8 means 1/8; 8:1 means 8. Ratio sets processing strength, not a fixed boost in dB or the fixed dB slope of a conventional compressor.

Quick reference: gain, monitoring and SC

Parameter	Range / choices	Notes
Input Gain	-24 to +24 dB	Level into the processing path.
Makeup	-36 to +36 dB	Default 0 dB; Wet level before Mix.
Mix	0 to 100%	Default 100%; included in effective dynamic gain.
Output Gain	-24 to +24 dB	Default 0 dB; final output level.
MONITOR	ALL / L / R or M / S	LR / MS audition; individual domain centred.
Lookahead	0 / 10 / 26 / 40 / 80 / 100 ms	Initial starting point 26 ms; new instances remember the last choice.
Oversampling	1x / 8x / 16x	Visible only at 0 ms; initially 8x.
SOURCE	INT / EXT	Default INT; EXT needs host routing.
KEY GAIN	-24 to +24 dB	Default 0 dB; adjustable only with EXT.
HPF	OFF / 20 to 500 Hz	Default OFF; filters only the detector key.
SC LISTEN	OFF / ON	Temporary audition; closing panel/editor switches it off.

Keep the gain controls distinct

Input changes the main input; Key Gain changes only the external detector; Makeup changes Wet before Mix; Output Gain changes the final output. Match loudness before comparing dynamics.

Upward / Dual troubleshooting

Symptom	First check
No audible boost after adjustment	Check that Single Ratio or Up Ratio is below 1:1, Mix is above 0%, and Bypass is off.
No upward effect in Dual	The detector must be above UP and below DOWN. Below UP, the original level is retained; above DOWN, only downward processing applies.
Faders set, but no processing	Do the boundaries coincide? This stops all dynamic processing. Separate them before comparing.
Loud parts suddenly stop being processed	Is a finite Range set in SINGLE? At or above it, dynamic gain returns to 0 dB. Select OFF if you want no upper cutoff.
Adjusting Up also changes Down	LINK between the two Ratios is on by default. Turn it off to adjust them independently.
LINK will not move further	The other Ratio may have reached its limit, stopping both controls. Reverse the adjustment to continue.
Main track level does not explain triggering	Check whether SOURCE is EXT or HPF is changing the detector signal. Threshold decisions follow the detector signal.

Check Dual upward processing alone

- 1 Keep other variables fixed**
 Select DUAL, turn Ratio LINK off and leave Down Ratio at 1:1. Keep Mix at 100% and all gain controls at 0 dB. Choose a suitable Lookahead.
- 2 Place the signal between the thresholds**
 For example, set UP to -50 dB and DOWN to -20 dB. Check that the detector level of a quiet passage lies mainly between them. With Ratio still at 1:1, there is no boost yet.
- 3 Increase upward strength from 1:1**
 Move Up Ratio through 1:2, 1:4 and 1:8 while listening and watching Boost / Mix and positive dynamic gain. Once confirmed, add Down processing and compare overall loudness again.

Troubleshooting

Symptom	First check
EXT has no effect	Is the host sidechain input/send enabled? Is SOURCE on EXT? Check KEY LEVEL: N/A means no external bus is available; -inf commonly means silence or no incoming signal.
Key Gain is disabled	This is normal with INT. Key Gain adjusts EXT only.
HPF makes little audible change	It does not directly cut the main signal's bass. Does the key contain low frequencies, and is compression active? Lower Mix also makes differences less audible.
HPF history does not follow every drag	History refreshes after release; wait for HPF UPDATING to clear. EXT Key Gain updates live.
I hear the sidechain source	Check whether SC LISTEN is on; turn it off for normal output.
One side only / narrowed image	Return MONITOR to ALL, then check LR / MS settings.
Match is unavailable	Play representative non-silent material to establish a usable loudness reading.
Host briefly realigns after a Lookahead change	Latency changes, so the host may realign compensation. Also consider latency while monitoring a live recording.
Nothing shown below -90 dB	This is the Display range, not a cutoff of the audio.

For installation, updates and plug-in scanning, see the supplied Windows / macOS installation guide.

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github.com/Ziqing-Gu/qq-super-compression